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Uncertainty principles for the windowed Hankel transform

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ABSTRACT
The aim of this paper is to prove some new uncertainty principles for the windowed Hankel transform. They include uncertainty principle for orthonormal sequence, local uncertainty principle, logarithmic uncertainty principle and Heisenberg-type uncertainty principle. As a side result, we obtain the Shapiro's dispersion theorem for the windowed Hankel transform.

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1. Introduction

The uncertainty principle is a very important theorem in harmonic analysis. It states that a non-zero function and its Fourier transform cannot be simultaneously sharply concentrated. As a classical uncertainty principle, the Heisenberg uncertainty principle has been extended to other transforms such as the linear canonical transform [1,2], the fractional Fourier transform [3,4] and the Dunkl transform [5]. The uncertainty principle for the Hankel transform was first proved by Bowie [6], then other uncertainty relations for the Hankel transform have been investigated [5,7–9].

The Hankel transform is found as a very useful mathematical tool in many fields of physics, geophysics, signal processing and other fields [10–12]. In a series of papers [13–16], various kinds of Hankel transform have been discussed in details. Ghobber and Omri [17] introduced the windowed Hankel transforms and described its basic properties. Baccar et al. [18] discussed the time-frequency analysis of localization operators the windowed transform in the Hankel setting. In this paper, our main aim is to introduce some uncertainty principles for the windowed Hankel transform.

To do so, we need to introduce some relevant contents.

The Hankel transform of order α is defined on $L^1_\alpha(\mathbb{R}_+)$ by [19]

$$\mathcal{H}_\alpha(f)(\lambda) = \int_0^{+\infty} f(t) j_\alpha(\lambda t)\, \mathrm{d}\gamma_\alpha(t), \tag{1.1}$$

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where $\alpha \geq -\frac{1}{2}$, $\mathrm{d}\gamma_\alpha$ the measure defined on $[0,+\infty)$ by

$$\mathrm{d}\gamma_\alpha(t) = \frac{t^{2\alpha+1}}{2^\alpha \Gamma(\alpha+1)}\,\mathrm{d}t, \tag{1.2}$$

where Γ is the gamma function and j_α is the Bessel function given by

$$j_\alpha(t) = \Gamma(\alpha+1)\sum_{n=0}^{\infty}\frac{(-1)^n}{n!\Gamma(n+\alpha+1)}\left(\frac{t}{2}\right)^{2n}.$$

The modified Bessel function $j_\alpha(t)$ has the following integral representation [20–22], for every $t \in \mathbb{C}$, we have

$$j_\alpha(t) = \begin{cases} \dfrac{2\Gamma(\alpha+1)}{\sqrt{\pi}\Gamma(\alpha+1/2)}\int_0^1 (1-x^2)^{\alpha-1/2}\cos(tx)\,\mathrm{d}x, & \text{if } \alpha > -\dfrac{1}{2}, \\ \cos z, & \text{if } \alpha = -\dfrac{1}{2}. \end{cases} \tag{1.3}$$

In particular, for every $z \in \mathbb{R}$, $|\ j_\alpha(z)\ | \leq 1$.

The integral representation (1.3) shows that for each $n \in \mathbb{N}$ and $t \in \mathbb{C}$, $|\ j_\alpha^{(n)}(t)\ | \leq e^{|Im(t)|}$. In particular, for each $n \in \mathbb{N}$ and $x \in \mathbb{R}$, $|\ j_\alpha^{(n)}(x)\ | \leq 1$.

$L^p_\alpha(\mathbb{R}_+)$ the space of measurable function f on $[0,+\infty)$ satisfying [8]

$$\|f\|_{L^p_\alpha(\mathbb{R}_+)} = \left(\int_0^{+\infty} |f(t)|^p\,\mathrm{d}\gamma_\alpha(t)\right)^{1/p} < +\infty, \quad \text{if } p \in [1,+\infty), \tag{1.4}$$

$$\|f\|_{L^\infty_\alpha(\mathbb{R}_+)} = \operatorname*{ess\ sup}_{t\in[0,+\infty)} |f(t)| < +\infty, \quad \text{if } p = +\infty. \tag{1.5}$$

Moreover, the Hankel transform satisfies the following inversion formula and Parseval equality [19]:

(1) Inversion formula: Let $f \in L^1_\alpha(\mathbb{R}_+)$ such that $\mathcal{H}_\alpha(f) \in L^1_\alpha(\mathbb{R}_+)$, then for every $t \in [0,+\infty)$ we have

$$f(t) = \int_0^{+\infty} \mathcal{H}_\alpha(f)(r) j_\alpha(rt)\,\mathrm{d}\gamma_\alpha(r).$$

(2) Parseval equality: The Hankel transform can be extended to an isometric isomorphism form $L^2_\alpha(\mathbb{R}_+)$ onto itself. Moreover, for every $f, g \in L^2_\alpha(\mathbb{R}_+)$, we have the following Parseval equality:

$$\int_0^{+\infty} f(t)\overline{g(t)}\,\mathrm{d}\gamma_\alpha(t) = \int_0^{+\infty} \mathcal{H}_\alpha(f)(r)\overline{\mathcal{H}_\alpha(g)(r)}\,\mathrm{d}\gamma_\alpha(r). \tag{1.6}$$

For every measurable subset $E \subseteq [0, +\infty)$, χ_E is the characteristic function of E such that [9]

$$\chi_E(t) = \begin{cases} 1, & t \in E, \\ 0, & t \in E^c. \end{cases} \tag{1.7}$$

The translation operator associated with the Hankel transform is defined by [23]

$$\tau_k^\alpha f(t) = \frac{\Gamma(\alpha+1)}{\Gamma(1/2)\Gamma(\alpha+(1/2))} \int_0^\pi f(\sqrt{t^2+k^2+2tk\cos\theta})(\sin\theta)^{2\alpha}\,\mathrm{d}\theta. \tag{1.8}$$

The operator τ_k^α can be also written by the formula

$$\tau_k^\alpha f(t) = \int_0^\infty f(x)K(k,t,x)\,\mathrm{d}\gamma_\alpha(x), \tag{1.9}$$

where $K(k,t,x)$ is the kernel given by

$$K(k,t,x) = \begin{cases} \dfrac{2\pi^{\alpha+1/2}\Gamma(\alpha+1)^2}{\Gamma(\alpha+(1/2))} \dfrac{\Delta(k,t,x)^{2\alpha-1}}{(ktx)^{2\alpha}}, & \text{if } |k-t| < x < k+t, \\ 0, & \text{otherwise,} \end{cases}$$

where $\Delta(k,t,x) = ((k+t)^2 - x^2)^{1/2}(x^2-(k-t)^2)^{1/2}$ is the area of the triangle with side length k, t, x.

The kernel $K(k,t,x)$ is symmetric in the variables k, t, x, and satisfy

$$\int_0^{+\infty} K(k,t,x)\,\mathrm{d}\gamma_\alpha(x) = 1.$$

For every $f \in L_\alpha^p(\mathbb{R}_+)$ and every $k \in \mathbb{R}_+$, the function $\tau_k^\alpha(f)$ belong to the space $L_\alpha^p(\mathbb{R}_+)$ and

$$\|\tau_k^\alpha(f)\|_{L_\alpha^p(\mathbb{R}_+)} \leq \|f\|_{L_\alpha^p(\mathbb{R}_+)}. \tag{1.10}$$

In particular, for every $k, t \geq 0$, we have

$$\tau_k^\alpha(f)(t) = \tau_t^\alpha(f)(k). \tag{1.11}$$

If $f \in L_\alpha^1(\mathbb{R}_+)$, then

$$\int_0^{+\infty} \tau_k^\alpha(f)(t)\,\mathrm{d}\gamma_\alpha(t) = \int_0^{+\infty} f(t)\,\mathrm{d}\gamma_\alpha(t). \tag{1.12}$$

The modulation operator associated with the Hankel transform is defined by

$$\mathcal{M}_s^\alpha g := \mathcal{H}_\alpha(\sqrt{\tau_s^\alpha|\mathcal{H}_\alpha(g)|^2}).$$

The convolution product associated with the Hankel transform is defined for two functions f and g by [8,24,25]

$$f\natural_\alpha g(t) = \int_0^{+\infty} f(r)\tau_t^\alpha(g)(r)\,\mathrm{d}\gamma_\alpha(r)$$

and

$$\mathcal{H}_\alpha(f\sharp_\alpha g) = \mathcal{H}_\alpha(f)\mathcal{H}_\alpha(g). \tag{1.13}$$

The paper is organized as follows: In Section 2, we present some preliminaries related to the windowed Hankel transform. Some different uncertainty principles associated with the windowed Hankel transform are provided in Section 3.

2. The windowed Hankel transform

Let $g \in L^2_\alpha(\mathbb{R}_+)$ and $s \in \mathbb{R}_+$, the modulation of g is defined by [17,18]

$$\mathcal{M}^\alpha_s g = \mathcal{H}_\alpha(\sqrt{\tau^\alpha_s|\mathcal{H}_\alpha(g)|^2}). \tag{2.1}$$

Then, for every $g \in L^2_\alpha(\mathbb{R}_+)$ and $s \in \mathbb{R}_+$, we obtained

$$\|\mathcal{M}^\alpha_s g\|_{L^2_\alpha(\mathbb{R}_+)} = \|g\|_{L^2_\alpha(\mathbb{R}_+)}. \tag{2.2}$$

For a non-zero window function $g \in L^2_\alpha(\mathbb{R}_+)$ and $(k,s) \in \mathbb{R}^2_+$, the function $g^\alpha_{k,s}$ defined by

$$g^\alpha_{k,s} = \tau^\alpha_k \mathcal{M}^\alpha_s g. \tag{2.3}$$

Now, for any function $f \in L^2_\alpha(\mathbb{R}_+)$, we define the windowed Hankel transform by [17,18]

$$W^\alpha_g(f)(k,s) = \int_0^{+\infty} f(t)\overline{g^\alpha_{k,s}(t)}\,\mathrm{d}\gamma_\alpha(t), \quad (k,s) \in \mathbb{R}^2_+, \tag{2.4}$$

which can be also written in the form

$$W^\alpha_g(f)(k,s) = f\sharp_\alpha\overline{\mathcal{M}^\alpha_s g(k)}. \tag{2.5}$$

Define the measure ν_α on $\mathbb{R}^*_+ \times \mathbb{R}_+$ by

$$\mathrm{d}\nu_\alpha(k,s) = \mathrm{d}\gamma_\alpha(k)\,\mathrm{d}\gamma_\alpha(s).$$

The windowed Hankel transform satisfies the following properties [17,18].

Proposition 2.1: *Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a non-zero window function, then we have*

(1) *The Cauchy–Schwarz inequality: For any $f \in L^2_\alpha(\mathbb{R}_+)$,*

$$\|W^\alpha_g(f)(k,s)\|_{L^\infty_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \le \|f\|_{L^2_\alpha(\mathbb{R}_+)}\|g\|_{L^2_\alpha(\mathbb{R}_+)}. \tag{2.6}$$

(2) *Plancherel's formula: For any $f \in L^2_\alpha(\mathbb{R}_+)$,*

$$\|W^\alpha_g(f)(k,s)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} = \|f\|_{L^2_\alpha(\mathbb{R}_+)}\|g\|_{L^2_\alpha(\mathbb{R}_+)}. \tag{2.7}$$

(3) *Orthogonality relation: For any $f, h \in L^2_\alpha(\mathbb{R}_+)$,*

$$\int_0^{+\infty}\int_0^{+\infty} W^\alpha_g(f)(k,s)\overline{W^\alpha_g(h)(k,s)}\,\mathrm{d}\nu_\alpha(k,s) = \|g\|^2_{L^2_\alpha(\mathbb{R}_+)}\int_0^{+\infty} f(t)\overline{h(t)}\,\mathrm{d}\gamma_\alpha(t). \tag{2.8}$$

(4) *Reproducing kernel Hilbert space: For any* $(k', s'); (k, s) \in \mathbb{R}_+ \times \mathbb{R}_+^*$,

$$H_g(k', s'; k, s) = \frac{1}{\|g\|^2_{L^2_\alpha(\mathbb{R}_+)}} g^\alpha_{k',s'} \sharp_\alpha \overline{\mathcal{M}^\alpha_s g(k)} = \frac{1}{\|g\|^2_{L^2_\alpha(\mathbb{R}_+)}} W^\alpha_g(g^\alpha_{k',s'})(k, s). \tag{2.9}$$

Furthermore, the kernel is pointwise bounded

$$| H_g(k', s'; k, s) | \leq 1, \quad \forall (k', s'); (k, s) \in \mathbb{R}_+ \times \mathbb{R}_+^*. \tag{2.10}$$

3. Uncertainty principle for the windowed Hankel transform

In this section, we obtain some uncertainty principles for the windowed Hankel transform.

We consider the following orthogonal projections [26,27]:

(1) O_g is the orthogonal projection from $L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}_+^*)$ onto $W^\alpha_g(L^2_\alpha(\mathbb{R}_+)$, R_ϕ is its range.
(2) O_E is the orthogonal projection on $L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}_+^*)$ defined by

$$O_E F = \chi_E F, \quad F \in L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}_+^*), \tag{3.1}$$

where $E \subset \mathbb{R}_+$, R_E is its range. We define $\|O_E O_g\| = sup\{\|O_E O_g(F)\|_{L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}_+^*)}, F \in L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}_+^*); \|F\|_{L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}_+^*)} = 1\}$.

Theorem 3.1: *Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a non-zero window function. For any subset $E \subset \mathbb{R}_+ \times \mathbb{R}_+^*$ and finite measure $\nu_\alpha(E) < +\infty$, then $O_E O_g$ is a Hilbert–Schmid operator and we have the following estimation:*

$$\|O_E O_g\|^2 \leq \nu_\alpha(E). \tag{3.2}$$

Proof: According to the paper [28], for every function $F \in L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}_+^*)$, the orthogonal projection O_g can be expressed as

$$O_g(F)(k, s) = \int_0^{+\infty} \int_0^{+\infty} F(k', s') H_g(k', s'; k, s) \, \mathrm{d}\nu_\alpha(k', s'),$$

where $H_g(k', s'; k, s)$ is defined by (2.9). By (3.1), we get

$$O_E O_g(F)(k, s) = \int_0^{+\infty} \int_0^{+\infty} \chi_E(k, s) F(k', s') H_g(k', s'; k, s) \, \mathrm{d}\nu_\alpha(k', s').$$

Using (1.4), (1.10), (2.9) and Fubini's theorem, we have

$$\|O_E O_g\|^2_{HS}$$
$$= \int_0^{+\infty} \int_0^{+\infty} \int_0^{+\infty} \int_0^{+\infty} |\chi_E(k, s)|^2 |H_g(k', s'; k, s)|^2 \, \mathrm{d}\nu_\alpha(k', s') \, \mathrm{d}\nu_\alpha(k, s)$$

$$= \frac{1}{\|g\|^2_{L^2_\alpha(\mathbb{R}_+)}} \int_E \int_E \left(\int_0^{+\infty} \int_0^{+\infty} \frac{1}{\|g\|^2_{L^2_\alpha(\mathbb{R}_+)}} \left| W_g^\alpha(g^\alpha_{k,s})(k',s') \right|^2 \, \mathrm{d}\nu_\alpha(k',s') \right) \mathrm{d}\nu_\alpha(k,s)$$

$$\leq \frac{\|g\|^2_{L^2_\alpha(\mathbb{R}_+)}}{\|g\|^2_{L^2_\alpha(\mathbb{R}_+)}} \nu_\alpha(E) = \nu_\alpha(E). \tag{3.3}$$

According to $O_E O_g$ is an integral operator with Hilbert–Schmidt kernel, we have $\|O_E O_g\|^2 \leq \|O_E O_g\|^2_{HS}$. Which completes the proof. ■

Theorem 3.2 (Uncertainty principle for orthonormal sequence): *Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a window function, $(\varphi_n)_{n\in\mathbb{N}}$ be an orthonormal sequence in $L^2_\alpha(\mathbb{R}_+)$ and any subset $E \subset \mathbb{R}_+ \times \mathbb{R}^*_+$. If $\nu_\alpha(E) < +\infty$, then for every $N \in \mathbb{N}^*$, we have*

$$\sum_{n=1}^{N} (1 - \|\chi_{E^c} W_g^\alpha(\varphi_n)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)}) \leq \nu_\alpha(E). \tag{3.4}$$

Proof: Let $(e_n)_{n\in\mathbb{N}}$ be an orthonormal basis of $L^2_\alpha(\mathbb{R}_+)$, since $O_E O_g$ is an integral operator with Hilbert–Schmidt kernel and satisfy relation (3.3), hence the positive operator $O_g O_E O_g$ satisfies

$$\sum_{n\in\mathbb{N}} \langle O_g O_E O_g e_n, e_n \rangle_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} = \|O_E O_g\|^2_{HS} \leq \nu_\alpha(E) < +\infty,$$

where $\langle \cdot, \cdot \rangle$ is the inner product in $L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}^*_+)$.

According to the paper [29], the positive operator $O_g O_E O_g$ is a trace class operator and

$$Tr(O_g O_E O_g) = \|O_E O_g\|^2_{HS} \leq \nu_\alpha(E).$$

Since $(\varphi_n)_{n\in\mathbb{N}}$ be an orthonormal sequence in $L^2_\alpha(\mathbb{R}_+)$, then by (2.8) we deduce that $(W_g^\alpha(\varphi_n))_{n\in\mathbb{N}}$ is an orthonormal sequence in $L^2_\alpha(\mathbb{R}_+ \times \mathbb{R}^*_+)$, thus

$$\begin{aligned} \sum_{n=1}^{N} \langle O_E W_g^\alpha(\varphi_n), W_g^\alpha(\varphi_n) \rangle_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} &= \sum_{n=1}^{N} \langle O_g O_E O_g W_g^\alpha(\varphi_n), W_g^\alpha(\varphi_n) \rangle_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \\ &\leq Tr(O_g O_E O_g). \end{aligned}$$

Hence, we obtain

$$\sum_{n=1}^{N} \langle O_E W_g^\alpha(\varphi_n), W_g^\alpha(\varphi_n) \rangle_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \leq \nu_\alpha(E).$$

On the other hand, by Cauchy–Schwarz inequality, we have, for every n, $1 \leq n \leq N$,

$$\begin{aligned} \langle O_E W_g^\alpha(\varphi_n), W_g^\alpha(\varphi_n) \rangle_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} &= 1 - \langle O_{E^c} W_g^\alpha(\varphi_n), W_g^\alpha(\varphi_n) \rangle_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \\ &\geq 1 - \|\chi_{E^c} W_g^\alpha(\varphi_n)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)}. \end{aligned}$$

Thus, we obtain

$$\sum_{n=1}^{N}(1-\|\chi_{E^c}W_g^\alpha(\varphi_n)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}) \leq \sum_{n=1}^{N}\langle O_E W_g^\alpha(\varphi_n), W_g^\alpha(\varphi_n)\rangle_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \leq \nu_\alpha(E).$$

■

Definition 3.3: Let $\delta > 0$ and $E \subset \mathbb{R}_+^* \times \mathbb{R}_+$ be a measurable subset. Let $f, g \in L^2_\alpha(\mathbb{R}_+)$ be two non-zero functions. We say that $W_g^\alpha(f)$ is δ-time-frequency concentrated on E, if

$$\|\chi_{E^c}W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \leq \delta\|W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}.$$

According to Theorem 3.2, we obtain the following result.

Proposition 3.4: *Let $0 < \delta < 1, r > 0$ and $B_r = \{(k,s) \in \mathbb{R}_+^* \times \mathbb{R}_+ \mid |(k,s)| \leq r\}$. Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a window function, $(\varphi_n)_{1\leq n\leq N}$ be an orthonormal sequence in $L^2_\alpha(\mathbb{R}_+)$. If $W_g^\alpha(\varphi_n)$ is δ-time-frequency concentrated in the ball B_r for every $1 \leq n \leq N$, then*

$$N \leq \frac{r^{4(\alpha+1)}}{2^{2(\alpha+1)}\Gamma(2\alpha+3)(1-\delta)}. \tag{3.5}$$

Proof: Using Theorem 3.2, we have

$$\sum_{n=1}^{N}(1-\|\chi_{B_r^c}W_g^\alpha(\varphi_n)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}) \leq \nu_\alpha(B_r). \tag{3.6}$$

For every $1 \leq n \leq N$, we obtain

$$\|\chi_{B_r^c}W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \leq \delta; \quad \nu_\alpha(B_r) = \frac{r^{4(\alpha+1)}}{2^{2(\alpha+1)}\Gamma(2\alpha+3)}.$$

Hence

$$N(1-\delta) \leq \frac{r^{4(\alpha+1)}}{2^{2(\alpha+1)}\Gamma(2\alpha+3)}.$$

■

Definition 3.5: Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a window function and $f \in L^2_\alpha(\mathbb{R}_+)$, we define the generalized pth time-frequency dispersion of the windowed Hankel transform by

$$\rho_p(W_g^\alpha(f)) = \left(\int_0^{+\infty}\int_0^{+\infty} |(k,s)|^p \, |W_g^\alpha(f)(k,s)|^2 \, \mathrm{d}\nu_\alpha(k,s)\right)^{1/p},$$

where $p > 0$ and $|(k,s)| = \sqrt{k^2+s^2}$.

The time dispersion of the windowed Hankel transform is defined by

$$\rho_{k,p}(W_g^\alpha(f)) = \left(\int_0^{+\infty}\int_0^{+\infty} |k|^p \, |W_g^\alpha(f)(k,s)|^2 \, \mathrm{d}\nu_\alpha(k,s)\right)^{1/p}.$$

The frequency dispersion of the windowed Hankel transform is defined by

$$\rho_{s,p}(W_g^\alpha(f)) = \left(\int_0^{+\infty}\int_0^{+\infty} |s|^p \, |W_g^\alpha(f)(k,s)|^2 \, \mathrm{d}\nu_\alpha(k,s)\right)^{1/p}.$$

Corollary 3.6: *Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a window function, $(\varphi_n)_{1\le n\le N}$ be an orthonormal sequence in $L^2_\alpha(\mathbb{R}_+)$ and $p > 0$. Fix $Y > 0$, if the sequence $\rho_p(W^\alpha_g(\varphi_n)) \le Y$ for every $1 \le n \le N$, then*

$$N \le \frac{2^{\frac{8}{p}(\alpha+1)-2\alpha-1} Y^{4(\alpha+1)}}{\Gamma(2\alpha+3)}.$$

Proof: Since, for every $r > 0$, we have

$$\begin{aligned}\|\chi_{B^c_r} W^\alpha_g(\varphi_n)\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} &= \int_{|(k,s)|\ge r}\int_{|(k,s)|\ge r} |(k,s)|^{-p}|(k,s)|^p|W^\alpha_g(\varphi_n)(k,s)|^2\,\mathrm{d}\nu_\alpha(k,s)\\ &\le r^{-p}\rho^p_p(W^\alpha_g(\varphi_n)) \le r^{-p}Y^p,\end{aligned}$$

if $r = 4^{1/p}Y$, we have

$$\|\chi_{B^c_{(4^{1/p}Y)}} W^\alpha_g(\varphi_n)\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \le \frac{1}{4}.$$

Hence for every $1 \le n \le N$, $W^\alpha_g(\varphi_n)$ is $\frac{1}{2}$-concentrated in the ball $B_{4^{1/p}Y}$, applying Proposition 3.4, we obtain

$$N \le \frac{2^{\frac{8}{p}(\alpha+1)-2\alpha-1} Y^{4(\alpha+1)}}{\Gamma(2\alpha+3)}. \qquad \blacksquare$$

Lemma 3.7: *Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a window function, $(\varphi_n)_{1\le n\le N}$ be an orthonormal sequence in $L^2_\alpha(\mathbb{R}_+)$ and $p > 0$. Then there exists $i_0 \in \mathbb{Z}$ such that*

$$\rho_p(W^\alpha_g(\varphi_n)) \ge 2^{i_0} \quad \forall n \in \mathbb{N}. \tag{3.7}$$

Proof: For each $i \in \mathbb{Z}$, we define

$$P_i = \{n \in \mathbb{N} | \rho_p(W^\alpha_g(\varphi_n)) \in [2^{i-1}, 2^i)\},$$

then for every $n \in P_i$, we have $\rho_p(W^\alpha_g(\varphi_n)) \le 2^i$.

Hence

$$\int_{|(k,s)|\ge 2^{i+2/p}}\int_{|(k,s)|\ge 2^{i+2/p}} |W^\alpha_g(\varphi_n)(k,s)|^2\,\mathrm{d}\nu_\alpha(k,s) \le \frac{1}{4}.$$

For every $n \in P_i$, $W^\alpha_g(\varphi_n)$ is $\frac{1}{2}$-concentrated in the ball $B_{2^{i+2/p}}$, according to Proposition 3.4, (1.1) and (2.4), we have P_i is finite and

$$N_i \le \frac{2^{\frac{8}{p}(\alpha+1)-2\alpha-1}}{\Gamma(2\alpha+3)} 2^{4(\alpha+1)i}, \tag{3.8}$$

where N_i is the number of elements in P_i. We see that P_i is empty for all $i < i_0$, so $\rho_p(W^\alpha_g(\varphi_n)) \ge 2^{i_0}$. $\blacksquare$

Theorem 3.8 (Shapiro's dispersion theorem for the windowed Hankel transform): *Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a window function and $(\varphi_n)_{1\leq n\leq N}$ be an orthonormal sequence in $L^2_\alpha(\mathbb{R}_+)$, then for $p > 0$ and $N \in \mathbb{N}$, we have*

$$\sum_{n=1}^{N} \rho_p^p(W_g^\alpha(\varphi_n)) \geq N^{1+\frac{p}{4(\alpha+1)}} \left(\frac{3\Gamma(2\alpha+3)}{2^{(8/p)(\alpha+1)+6\alpha+8}}\right)^{p/4(\alpha+1)}.$$

Proof: Let $m \in \mathbb{Z}$, then according to (3.6), for every $m > i_0$, the number of element in $\bigcup_{i=i_0}^{m} P_i$ is less than $Q2^{4(\alpha+1)m}$, where

$$Q = \frac{2^{\frac{8}{p}(\alpha+1)+2\alpha+3}}{3\Gamma(2\alpha+3)},$$

is a constant that does not depend on m.

Now if $N > 2Q2^{4i_0(\alpha+1)}$, then we can choose an integer $m > i_0$ such that

$$2Q2^{4(m-1)(\alpha+1)} < N \leq 2Q2^{4m(\alpha+1)}.$$

Therefore, at least half of $1, \ldots, N$ do not belong to $\bigcup_{i=i_0}^{m-1} P_i$ and we have

$$\sum_{n=1}^{N} \rho_p^p(W_g^\alpha(\varphi_n)) \geq \frac{N}{2}2^{(m-1)p} \geq N^{1+p/4(\alpha+1)} \left(\frac{3\Gamma(2\alpha+3)}{2^{(12/p)(\alpha+1)+6\alpha+5}}\right)^{p/4(\alpha+1)}.$$

Finally, if $N \leq 2Q2^{4i_0(\alpha+1)}$, then we have

$$\sum_{n=1}^{N} \rho_p^p(W_g^\alpha(\varphi_n)) \geq N2^{(m-1)p} \geq N^{1+p/4(\alpha+1)} \left(\frac{3\Gamma(2\alpha+3)}{2^{(8/p)(\alpha+1)+6\alpha+8}}\right)^{p/4(\alpha+1)}.$$

This completes the proof. ■

Theorem 3.9: *Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a window function such that $\|g\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} = 1$. Suppose that $\|f\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} = 1$, then for $E \subset \mathbb{R}_+ \times \mathbb{R}_+^*$ and $\eta \geq 0$ such that*

$$\int_E\int_E \mid W_g^\alpha(f)(k,s) \mid^2 \, \mathrm{d}\nu_\alpha(k,s) \geq 1-\eta,$$

we have

$$\nu_\alpha(E) \geq 1-\eta.$$

Proof: According to (2.6), we have

$$1-\eta \leq \int_E\int_E \mid W_g^\alpha(f)(k,s) \mid^2 \, \mathrm{d}\nu_\alpha(k,s) \leq \| W_g^\alpha(f)\|_{L^\infty_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}\nu_\alpha(E) \leq \nu_\alpha(E).$$

■

Proposition 3.10: *Let $g \in L^2_\alpha(\mathbb{R}_+)$ be a window function such that $\|g\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} = 1$. Then for every function $f \in L^2_\alpha(\mathbb{R}_+)$ and $E \subset \mathbb{R}_+ \times \mathbb{R}^*_+$ such that $\nu_\alpha(E) < 1$, then*

$$\|\chi_{E^c} W^\alpha_g(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \geq \sqrt{1-\nu_\alpha(E)}\|f\|_{L^2_\alpha(\mathbb{R}_+)}.$$

Proof: By (2.6), we have

$$\begin{aligned}\|W^\alpha_g(f)\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} &= \|\chi_E W^\alpha_g(f) + \chi_{E^c} W^\alpha_g(f)\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)}\\ &\leq \|\chi_E W^\alpha_g(f)\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} + \|\chi_{E^c} W^\alpha_g(f)\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)}\\ &\leq \nu_\alpha(E)\|W^\alpha_g(f)\|^2_{L^\infty_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} + \|\chi_{E^c} W^\alpha_g(f)\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)}\\ &\leq \nu_\alpha(E)\|g\|^2_{L^2_\alpha(\mathbb{R}_+)}\|f\|^2_{L^2_\alpha(\mathbb{R}_+)} + \|\chi_{E^c} W^\alpha_g(f)\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)}.\end{aligned}$$

Hence

$$\|\chi_{E^c} W^\alpha_g(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \geq \sqrt{1-\nu_\alpha(E)}\|f\|_{L^2_\alpha(\mathbb{R}_+)}. \qquad \blacksquare$$

Theorem 3.11 (Local uncertainty inequality for the windowed Hankel transform): *Let x be a real number such that $0 < x < \alpha + 1$, $t \geq 0$, then for every non-zero function $f \in L^2_\alpha(\mathbb{R}_+)$ and for every measurable subset $E \subset \mathbb{R}_+ \times \mathbb{R}^*_+$ such that $0 < \nu_\alpha(E) < +\infty$, we have*

$$\begin{aligned}\|\chi_E W^\alpha_g(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} &\leq \left(1 + \frac{x^2}{(\nu_\alpha(E))^{1/2}(\alpha+1-x)^2}\right)\left(\frac{x^2 2^{\alpha+1}\Gamma(\alpha+1)}{\nu_\alpha(E)(\alpha+1-x)}\right)^{-x/(\alpha+1)}\\ &\quad \times \|t^x f\|_{L^2_\alpha(\mathbb{R}_+)}\|t^x g\|_{L^2_\alpha(\mathbb{R}_+)}. \end{aligned} \tag{3.9}$$

Proof: For every $a > 0$, according to Proposition 3.10, we have

$$\begin{aligned}\|\chi_E W^\alpha_g(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} &\leq (\nu_\alpha(E))^{1/2}\|\chi_{[0,a)} f\|_{L^2_\alpha(\mathbb{R}_+)}\|\chi_{[0,a)} g\|_{L^2_\alpha(\mathbb{R}_+)}\\ &\quad + \|W^\alpha_g(\chi_{[a,+\infty)} f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)}.\end{aligned}$$

However, by Hölder's inequality, we have

$$\|\chi_{[0,a)} f\|_{L^2_\alpha(\mathbb{R}_+)} \leq \|t^x f\|_{L^2_\alpha(\mathbb{R}_+)}\|\chi_{[0,a)} t^{-x}\|_{L^2_\alpha(\mathbb{R}_+)}.$$

According to (1.2), (1.4) and (1.7), we obtain

$$\begin{aligned}\|\chi_{[0,a)} t^{-x}\|_{L^2_\alpha(\mathbb{R}_+)} &= \left(\int_0^a (t^{-x})^2\, \mathrm{d}\gamma_\alpha(t)\right)^{1/2}\\ &= \left(\int_0^a t^{-2x} \cdot \frac{t^{2\alpha+1}}{2^\alpha \Gamma(\alpha+1)}\, \mathrm{d}t\right)^{1/2}\\ &= \frac{a^{\alpha+1-x}}{(2^{\alpha+1}\Gamma(\alpha+1)(\alpha+1-x))^{1/2}}.\end{aligned}$$

Hence

$$\|\chi_{[0,a)}f\|_{L^2_\alpha(\mathbb{R}_+)} \leq \|t^x f\|_{L^2_\alpha(\mathbb{R}_+)} \frac{a^{\alpha+1-x}}{(2^{\alpha+1}\Gamma(\alpha+1)(\alpha+1-x))^{1/2}}.$$

By (2.7), we have

$$\begin{aligned}\|W^\alpha_g(\chi_{[a,+\infty)}f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} &= \|\chi_{[a,+\infty)}f\|_{L^2_\alpha(\mathbb{R}_+)}\|\chi_{[a,+\infty)}g\|_{L^2_\alpha(\mathbb{R}_+)} \\ &\leq \|t^x f\|_{L^2_\alpha(\mathbb{R}_+)}\|\chi_{[a,+\infty)}t^{-x}\|_{L^\infty_\alpha(\mathbb{R}_+)} \\ &\quad\times \|t^x g\|_{L^2_\alpha(\mathbb{R}_+)}\|\chi_{[a,+\infty)}t^{-x}\|_{L^\infty_\alpha(\mathbb{R}_+)} \\ &= a^{-2x}\|t^x f\|_{L^2_\alpha(\mathbb{R}_+)}\|t^x g\|_{L^2_\alpha(\mathbb{R}_+)}.\end{aligned}$$

Hence

$$\begin{aligned}\|\chi_E W^\alpha_g(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \leq &\left(a^{-2x} + \frac{(\nu_\alpha(E))^{1/2}}{2^{\alpha+1}\Gamma(\alpha+1)(\alpha+1-x)}a^{2\alpha+2-2x}\right) \\ &\times \|t^x f\|_{L^2_\alpha(\mathbb{R}_+)}\|t^x g\|_{L^2_\alpha(\mathbb{R}_+)},\end{aligned} \tag{3.10}$$

in particular, inequality (3.10) holds for

$$a = \left(\frac{x^2 2^{\alpha+1}\Gamma(\alpha+1)}{\nu_\alpha(E)(\alpha+1-x)}\right)^{1/2(\alpha+1)}.$$

Hence

$$\begin{aligned}\|\chi_E W^\alpha_g(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}^*_+)} \leq &\left(1 + \frac{x^2}{(\nu_\alpha(E))^{1/2}(\alpha+1-x)^2}\right)\left(\frac{x^2 2^{\alpha+1}\Gamma(\alpha+1)}{\nu_\alpha(E)(\alpha+1-x)}\right)^{-x/(\alpha+1)} \\ &\times \|t^x f\|_{L^2_\alpha(\mathbb{R}_+)}\|t^x g\|_{L^2_\alpha(\mathbb{R}_+)}.\end{aligned}$$ ■

According to the following logarithmic uncertainty principle for the Hankel transform [8], we obtain the logarithmic uncertainty principle for the windowed Hankel transform.

Proposition 3.12: *For every $f \in L^2_\alpha(\mathbb{R}_+)$, the following inequality holds:*

$$\begin{aligned}&\int_0^{+\infty} \ln(t)|f(t)|^2\,\mathrm{d}\gamma_\alpha(t) + \int_0^{+\infty} \ln(\lambda)|\mathcal{H}_\alpha(f)(\lambda)|^2\,\mathrm{d}\gamma_\alpha(\lambda) \\ &\quad\geq \left(\psi\left(\frac{\alpha+1}{2}\right) + \ln(2)\right)\int_0^{+\infty} |f(t)|^2\,\mathrm{d}\gamma_\alpha(t),\end{aligned} \tag{3.11}$$

where ψ denotes the logarithmic derivative of the Gamma function Γ [20,21].

Now we arrive at the following important result.

Theorem 3.13 (Logarithmic uncertainty principle for the windowed Hankel transform): *For every $g \in L^2_\alpha(\mathbb{R}_+)$ and $f \in L^1_\alpha(\mathbb{R}_+) \bigcap L^2_\alpha(\mathbb{R}_+)$ we have the following inequality:*

$$\int_0^{+\infty}\int_0^{+\infty} \ln(k)|W_g^\alpha(f)(k,s)|^2\,\mathrm{d}\nu_\alpha(k,s) + \int_0^{+\infty}\int_0^{+\infty} \ln(s)|W_f^\alpha(g)(k,s)|^2\,\mathrm{d}\nu_\alpha(k,s)$$
$$\geq \left(\psi\left(\frac{\alpha+1}{2}\right)+\ln(2)\right)\|f\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}\|g\|^2_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}, \tag{3.12}$$

where ψ denotes the logarithmic derivative of the Gamma function Γ.

Proof: We can replace the logarithmic uncertainty principle for the Hankel transform to the function $W_g^\alpha(f)(k,s)$, then we have

$$\int_0^{+\infty} \ln(k)|W_g^\alpha(f)(k,s)|^2\,\mathrm{d}\gamma_\alpha(k) + \int_0^{+\infty} \ln(\lambda)|\mathcal{H}_\alpha(W_g^\alpha(f)(\cdot,s))(\lambda)|^2\,\mathrm{d}\gamma_\alpha(\lambda)$$
$$\geq \left(\psi\left(\frac{\alpha+1}{2}\right)+\ln(2)\right)\int_0^{+\infty} |W_g^\alpha(f)(k,s)|^2\,\mathrm{d}\gamma_\alpha(k).$$

Integrating both sides with respect to s, we have

$$\int_0^{+\infty}\int_0^{+\infty} \ln(k)|W_g^\alpha(f)(k,s)|^2\,\mathrm{d}\nu_\alpha(k,s)$$
$$+\int_0^{+\infty}\int_0^{+\infty} \ln(\lambda)|\mathcal{H}_\alpha(W_g^\alpha(f)(\cdot,s))(\lambda)|^2\,\mathrm{d}\nu_\alpha(\lambda,s)$$
$$\geq \left(\psi\left(\frac{\alpha+1}{2}\right)+\ln(2)\right)\int_0^{+\infty}\int_0^{+\infty} |W_g^\alpha(f)(k,s)|^2\,\mathrm{d}\nu_\alpha(k,s). \tag{3.13}$$

By (1.13) and (2.5), we obtain

$$\int_0^{+\infty}\int_0^{+\infty} \ln(\lambda)|\mathcal{H}_\alpha(W_g^\alpha(f)(\cdot,s))(\lambda)|^2\,\mathrm{d}\nu_\alpha(\lambda,s)$$
$$= \|g\|^2_{L^2_\alpha(\mathbb{R}_+)}\int_0^{+\infty} \ln(\lambda)|\mathcal{H}_\alpha(f)(\lambda)|^2\,\mathrm{d}\gamma_\alpha(\lambda). \tag{3.14}$$

Hence, by (2.7), we have

$$\int_0^{+\infty}\int_0^{+\infty} \ln(k)|W_g^\alpha(f)(k,s)|^2\,\mathrm{d}\nu_\alpha(k,s) + \|g\|^2_{L^2_\alpha(\mathbb{R}_+)}\int_0^{+\infty} \ln(\lambda)|\mathcal{H}_\alpha(f)(\lambda)|^2\,\mathrm{d}\gamma_\alpha(\lambda)$$
$$\geq \left(\psi(\frac{\alpha+1}{2})+\ln(2)\right)\|f\|^2_{L^2_\alpha(\mathbb{R}_+)}\|g\|^2_{L^2_\alpha(\mathbb{R}_+)}. \tag{3.15}$$

On the other hand, by (1.6) and (1.13), we have

$$\int_0^{+\infty}\int_0^{+\infty} \ln(s)|W_f^\alpha(g)(k,s)|^2\,\mathrm{d}\nu_\alpha(k,s)$$
$$= \int_0^{+\infty} \ln(s)\left[\int_0^{+\infty} |\mathcal{H}_\alpha[W_f^\alpha(g)(\cdot,s)](\omega)|^2\,\mathrm{d}\gamma_\alpha(\omega)\right]\mathrm{d}\gamma_\alpha(s)$$

$$= \int_0^{+\infty} \ln(s)\Big[\int_0^{+\infty} |\mathcal{H}_\alpha(g)(\omega)|^2 \tau_s^\alpha \times |\mathcal{H}_\alpha(f)(\omega)|^2\, d\gamma_\alpha(\omega)\Big]\, d\gamma_\alpha(s)$$
$$= \int_0^{+\infty} |\mathcal{H}_\alpha(g)(\omega)|^2 \times \left[\int_0^{+\infty} \ln(s)\tau_\omega^\alpha |\mathcal{H}_\alpha(f)(s)|^2\, d\gamma_\alpha(s)\right] d\gamma_\alpha(\omega).$$

According to (1.12), we have

$$\int_0^{+\infty} \ln(s)\tau_\omega^\alpha |\mathcal{H}_\alpha(f)|^2(s)\, d\gamma_\alpha(s) = \int_0^{+\infty} \ln(s)|\mathcal{H}_\alpha(f)|^2(s)\, d\gamma_\alpha(s).$$

Hence

$$\int_0^{+\infty}\int_0^{+\infty} \ln(s)|W_f^\alpha(g)(k,s)|^2\, d\nu_\alpha(k,s)$$
$$= \int_0^{+\infty} |\mathcal{H}_\alpha(g)(\omega)|^2\, d\gamma_\alpha(\omega) \int_0^{+\infty} \ln(s)|\mathcal{H}_\alpha(f)|^2(s)\, d\gamma_\alpha(s)$$
$$= \|g\|^2_{L^2_\alpha(\mathbb{R}_+)} \int_0^{+\infty} \ln(s)|\mathcal{H}_\alpha(f)(s)|^2\, d\gamma_\alpha(s). \tag{3.16}$$

Which completes the proof. ∎

In the paper [30], Soltani gave explicitly the constant b in the case $c \ge 1$ and $d \ge 1$, more precisely he established the following Heisenberg-type inequalities for the Hankel transform.

Proposition 3.14: *Assume $c \ge 1$ and $d \ge 1$, then for every function $f \in L^2_\alpha(\mathbb{R}_+)$, we have*

$$\|t^d f\|^{c/(c+d)}_{L^2_\alpha(\mathbb{R}_+)} \|\lambda^c \mathcal{H}_\alpha(f)\|^{d/(c+d)}_{L^2_\alpha(\mathbb{R}_+)} \ge (\alpha+1)^{cd/(c+d)} \|f\|_{L^2_\alpha(\mathbb{R}_+)}, \tag{3.17}$$

with equality if and only if $c = d = 1$ and $f(t) = pe^{-qt^2/2}$ for some $p \in \mathbb{C}$ and $q > 0$.

In the paper [17], the authors gave the following Heisenberg-type uncertainty inequality for the windowed Hankel transform of magnitude $c \ge 1$.

Proposition 3.15: *Let $c \ge 1$, then for every function $f, g \in L^2_\alpha(\mathbb{R}_+)$,*

$$\|k^c W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \|s^c W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \ge (\alpha+1)^c \|g\|^2_{L^2_\alpha(\mathbb{R}_+)} \|f\|^2_{L^2_\alpha(\mathbb{R}_+)}. \tag{3.18}$$

When $c = 1$, the uncertainty inequality become the following uncertainty inequality:

$$\|k W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \|s W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \ge (\alpha+1) \|g\|^2_{L^2_\alpha(\mathbb{R}_+)} \|f\|^2_{L^2_\alpha(\mathbb{R}_+)}. \tag{3.19}$$

Now, we investigate the Heisenberg-type uncertainty inequality for the windowed Hankel transform.

Theorem 3.16 (Heisenberg-type uncertainty inequality for the windowed Hankel transform): *Assume $c, d \geq 1$ and let $g \in L^2_\alpha(\mathbb{R}_+)$ be a non-zero window function. For every non-zero function $f \in L^2_\alpha(\mathbb{R}_+)$, then there exits a constant $b = (\alpha, c, d) > 0$, such that*

$$\|k^c W_g^\alpha(f)\|^{d/(c+d)}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \|s^d W_g^\alpha(f)\|^{c/(c+d)}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \geq (\alpha+1)^{cd/(c+d)} \|g\|_{L^2_\alpha(\mathbb{R}_+)} \|f\|_{L^2_\alpha(\mathbb{R}_+)}. \tag{3.20}$$

Proof: Let $f, g \in L^2_\alpha(\mathbb{R}_+)$ be two non-zero functions such that

$$\|k^c W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}; \|s^d W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} < \infty.$$

Then, for $c > 1$, we have

$$\begin{aligned} &\|k^c W_g^\alpha(f)\|^{1/c}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \|W_g^\alpha(f)\|^{1/c'}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \\ &\quad = \|k^2 |W_g^\alpha(f)|^{2/c}\|^{1/2}_{L^c_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \||W_g^\alpha(f)|^{2/c'}\|^{1/2}_{L^{c'}_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}, \end{aligned}$$

where c' is defined as usual by $1/c + 1/c' = 1$.

By Hölder's inequality, we obtain

$$\|k^c W_g^\alpha(f)\|^{1/c}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \|W_g^\alpha(f)\|^{1/c'}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \geq \|k W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}.$$

Thus, for all $c \geq 1$, we have

$$\|k^c W_g^\alpha(f)\|^{1/c}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \geq \frac{\|k W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}}{\|W_g^\alpha(f)\|^{1-1/c}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}}, \tag{3.21}$$

with equality if $c = 1$.

In the same way, we obtain, for $d \geq 1$,

$$\|s^d W_g^\alpha(f)\|^{1/d}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \geq \frac{\|s W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}}{\|W_g^\alpha(f)\|^{1-1/d}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}}, \tag{3.22}$$

with equality if $d = 1$.

According to (3.21) and (3.22), for all $c, d \geq 1$, we obtain

$$\begin{aligned} &\|k^c W_g^\alpha(f)\|^{d/(c+d)}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \|s^d W_g^\alpha(f)\|^{c/(c+d)}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \\ &\quad \geq \left(\frac{\|k W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)} \|s W_g^\alpha(f)\|_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}}{\|W_g^\alpha(f)\|^{2-1/c-1/d}_{L^2_\alpha(\mathbb{R}_+\times\mathbb{R}_+^*)}} \right)^{cd/(c+d)}, \end{aligned} \tag{3.23}$$

with equality if $c = d = 1$.

By (2.7) and (3.19), we obtain the desired result. ∎

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